

A Comparative Patch-wise analysis of multiclass weed detection using Sorghum dataset

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Abstract—This project presents a comprehensive deep learning approach for image recognition and classification using multiple light weight algorithms (ResNet50, InceptionV3, MobileNetV3, Xception, VGG16, and EfficientNet) through transfer learning technique. The ADAM optimizer and categorical cross-entropy loss, at the same time adaptive learning rates are used to prevent over-fitting of the model. Performance metrics include accuracy, confusion matrices, and classification reports for evaluation. Then patch-wise classification method is realized, where the input images are divided into smaller grid-based patches whose size varies from 64x64 to 8x8 and class probabilities are predicted for each grid. If a particular class is indicated, that region is marked with the respective colour. If multiple class is detected, the grid is again reduced to half, and the same procedure is repeated. This repeated grid size reduction gives more detailed and localized patch-wise analysis, capturing the results more precisely.

Index Terms—Deep learning, Image recognition, Multi-class image classification, ResNet50, InceptionV3, MobileNetV3, Xception, VGG16, EfficientNet-B0, , feature extraction, data enhancement, ADAM optimizer, categorical cross-entropy loss, patch-wise classification, grid size reduction, class probability, class prediction.

I. INTRODUCTION

The project initially focuses on the transfer learning process to a variety of state-of-the-art deep learning models, including ResNet50, MobileNetV3, and EfficientNet, which are used as feature extractors for image classification services. Custom fully connected layers are used for object classification, while InceptionV3, Xception, and VGG16 are optimized for flexibility and accuracy in garbage classification.

Then the project focuses on development of an effective image classification and detection algorithm for SORGHUM weed datasets using two advanced convolution neural networks (CNNs): VGG16 and Xception, aim to combine the

TABLE I
TRAINABLE AND NON-TRAINABLE PARAMETERS FOR VARIOUS ARCHITECTURES IN WEED IDENTIFICATION AND CLASSIFICATION

Architecture	Trainable Parameters	Non-Trainable Parameters
Resnet 50	23,587,712	53,184
Inception V3	21,802,784	85,696
MobileNet V3	4,206,984	10,112
Xception	22,910,480	78,560
Efficientnet B0	4,049,928	27,232
VGG16	138,357,544	0

strengths of both architectures for classification accuracy has improved in grasslands. The method involves preprocessing of image data, including techniques for model enhancement, and transfer learning to be used to optimize pre-trained models. The system is trained using categorical cross-entropy loss and evaluated over various performance metrics including confusion matrices and classification reports to ensure correct identification of target classes.

A patch-wise classification technique is introduced, where the input images are divided into mesh-based patches. Each patch is segmented, and if a particular class in the patch is recognized, that region is marked with a particular colour. If multiple classes are recognized, the patch size is reduced by half, and the classification process is repeated. This repeated process provides localized and accurate classes in an input image. Deep learning techniques such as early dropout is used to prevent overfitting of the model. The performance of the model is evaluated on the basis of accuracy, confusion matrices, and classification reports, class-wise area contribution, class probability, patch accuracy

which are important in evaluation of quality of the patch-wise classification method.

II. RELATED WORKS

An analytical method detecting Vegetable garden weeds using deep Learning and image processing in deployed in one of the related sources. Through the initial training Vegetables see a pattern, the system can then select weeds as dropping material from the vegetable border boxes. This is the solution helps simplifying and improving weed identification accuracy, and reduce character complexity and various types of weed species[1]. Another research examines advanced technologies for weeds is used in specific areas, e.g A camera-guided hoe and a robot lawn. These have a precise and sustainable weed control in agriculture. The focus is on reduction of herbicide use, increased biodiversity, and a robust system development along with farmer support. The goal is to achieve environmental goals and improve agricultural practices[2]. The paper shows a different CVDL-WDC method of classifying the plant accurately and weed also on the field. This uses WF preprocessing, multiscale RCNN object detection, ELM based classification and FFO parameter optimization for optimal training results. The learning improves optimization of the ELM model using FFO. This system gives the best efficiency than the old methods and produces better results are yield. This technique can be easily implemented in IoT devices and phones. This feature can use a wide range of devices, becoming a positive adaptive technique and can also be used for future developments.[3]

Other approach bring a new style to the picture Processing by indulging more advanced The methods of operation. This uses the noun suffix PoolFormer, in conjunction with Efficient Channel Attention (ECA). Extraction of materials. In addition, there is a trap It was used called the Feature Alignment Pyramid network (FaPN), modify loss functions for the better performance. This method received high marks which is 87.45% for the convergence on the combination (MIoU) and 96.82% for pixel accuracy. (MPA), indicating good results receipt[4]. Another paper improves the existing model, YOLO, InceptionV3 integration included. This development is a coincidence Initialization modules, in the YOLOv3 system. in Doing this allows the model to exclude features more effectively From drawings on different scales, a. Accuracy improved. Hence the precedent modified can better understand and recognize objects note their parts in the figures on more effective scales[5].

Some more techniques uses the MobilenetV3 algorithm Proper classification of plant diseases. For the next step enhancement, the paper introduces a convolution the aggregator and the opposite side. such as MobilenetV3 Previously trained in ImageNet, he already has a high Because of its light weight, the accuracy rate is (95pattern. So the use of paper, To build MobileNetV3 performance and

accuracy Advances in the classification of plant diseases, . which can be used on mobile platforms[6].

III. METHODOLOGY

A. Dataset

TABLE II
CLASS DISTRIBUTION OF WEED IDENTIFICATION DATASET

Class	Number of Samples	Percentage
Sorghum Saplings (Class 0)	1,404	32.5%
Grasses (Class 1)	1,467	34.1%
Broad-leaf Weeds (Class 2)	1,441	33.4%

In this project, we have used weed classification and segmentation dataset called SORGHUM, containing 4,312 image samples. There are 2 divisions under classification section namely, 'Test' and 'TrainNValidate'. Class 0 consists of 1,404 samples of Sorghum samples; approximately 32.5% of the total number of plant samples in the dataset. Class 1 contains 1,467 samples of Grass weeds approximately 34.1% of the dataset. Class 2 contains 1,441 samples of Broad-leaf Weeds approximately 33.4% of the total number of samples in the dataset.

All images of classification file are resized to 224 x 224 pixels to make it easier for the deep learning (DL) architectures, which are being used for realization if the project. The dataset will be equally evaluated by splitting it into training, validation, and testing using a 90:10 ratio: 90 percent training and validation, 10 percent testing. This provides a rigorous structure to classify crop and weed species through a neural network, allowing ample samples for training and variability in the training dataset from the other training dataset, while also allowing a rigorous testing of the training dataset following sample-by-sample layout and validation.

Our approach will be focused on the identification and classification of weeds, especially sorghum (class 0), using different light weight deep learning architectures mentioned above. It's first segment involves deployment of deep learning in SORGHUM dataset. The dataset will be divided into training&validation and testing sets, typically with a ratio of 10 percent for testing, 90 percent for the rest.

B. Working Flow

Here are Mathematical Breakdown of Patch-Wise Classification and Optimization Workflow:

1) *Patch-Wise Classification*: Given an image I with dimensions $H \times W \times C$, we divide it into smaller patches, each of size $p \times p$. The total number of patches is given by:

$$N_p = \left(\frac{H}{p}\right) \times \left(\frac{W}{p}\right)$$

Each patch P_i is passed through the feature extractor part of the model (e.g., ResNet50, InceptionV3) and classified.

2) *Transfer Learning and Feature Extraction*: Using transfer learning, the pretrained models (e.g., ResNet50, MobileNetV3) act as feature extractors. This is represented as:

$$\text{Features}(P_i) = \phi(P_i; \theta_{\text{pretrained}})$$

where ϕ is the pretrained model with frozen convolutional layers, and $\theta_{\text{pretrained}}$ represents the pretrained parameters.

3) *Classification Head*: After extracting features for each patch, the classification layer maps the extracted features to one of the classes C_j , from a set of classes $\{C_1, C_2, \dots, C_k\}$:

$$\hat{C}_i = \text{Softmax}(W \cdot \text{Features}(P_i) + b)$$

where W and b are the learned weights in the classification head, and $\hat{C}_i$ is the predicted class for patch P_i .

4) *Loss Function*: The categorical cross-entropy loss function L for each patch is written as:

$$L = - \sum_{i=1}^{N_p} \sum_{j=1}^k y_{ij} \log(\hat{y}_{ij})$$

where y_{ij} is the true label, and $\hat{y}_{ij}$ is the predicted probability for class j for patch P_i .

5) *Adam Optimizer*: The Adam optimizer updates the weights using gradients from the loss function with adaptive learning rates. For each parameter θ_t , Adam updates it as follows:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla_{\theta_t} L_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla_{\theta_t} L_t)^2$$

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t} + \epsilon}$$

where m_t and v_t are the first and second moments, η is the learning rate, and ϵ is a small constant for numerical stability.

6) *Confusion Matrix*: The confusion matrix evaluates model performance across all classes, calculating true positives, false positives, etc.:

$$\text{Confusion Matrix} = [\text{True Positive}, \text{False Positive}, \dots]$$

7) *Accuracy*: The accuracy is computed as:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total predictions}}$$

8) *Classification Report*: The classification report provides precision, recall, and F1-score for each class:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The figure shows the workflow for waste identification and classification in fields. It starts with data sets divided into training, validation, and testing. These algorithms are then used to train and test various parts of the system, including grassland contribution analysis, grassland segmentation, and strip segmentation. The process culminates in model ensembling, where multiple models are combined to improve performance. The results are evaluated based on metrics such as accuracy, precision, recall and F1-score. The system is designed to identify and classify weeds in images, which can be useful for precision agricultural applications such as targeted herbicides.

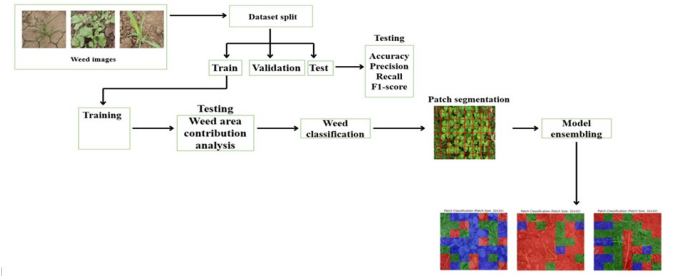


Fig. 1. Methodology of block diagram

Then, the method selects pre-trained models using transfer learning. For each architecture—VGG16, InceptionV3, MobileNetV3, EfficientNet, and Xception—pre-trained weights are generated from ImageNet. The first convolutional layers responsible for feature extraction will be frozen to preserve their known features, while the later layers will be frozen for fine-tuning. A custom output layer will be added to match the number of grass classes the dataset is consistent.

The training algorithm will use categorical cross-entropy as the loss function for multi-class classification, ADAM optimizer. During the training process, each model will go through a fine-tuning process to optimize its performance for a reasonable number of times. Checking for loss of validation accuracy will be performed to avoid overfitting, including the use of callbacks such as early stops and model checkpoints to preserve the best performing models.

After training, each model will be tested in a test setup, and performance metrics will be collected to compare the effectiveness of different designs and processing functions compare visualization techniques including confusion matrices and ROC curves will help develop model performance in the analysis

C. Model Ensembling

1) *VGG16*: VGG16 has 16-layers with sequential convolution and fully connected layers. As shown in Fig. 3, list consists of several convolutional layers followed by ReLU activation to capture low-level, mid-level, and high-level features as the input image is traversed through the network. It uses small 3x3 filters for these layers but instead of applying one per layer, we stack multiple convolutional with the goal of increasing depth and at same time feature extraction. The deeper the layers go, more complicated patterns are detected which is important for inspection of detailed lines on coconut tree diseases. The last fully connected layers which converts high level features into one of the five classes in this study.

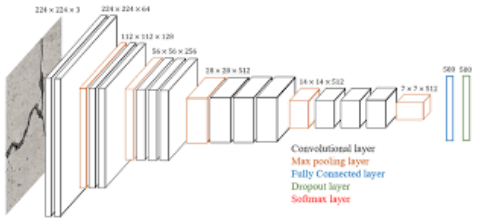


Fig. 2. VGG16 Architecture

2) *Xception*: The Xception model is based on the principle of "extreme" Inception, as illustrated in Fig. 4. Unlike the traditional Inception module, Xception uses depth-wise separable convolutions throughout its architecture, which significantly reduces the number of parameters while maintaining high efficiency. This helps the model focus more on important spatial information by learning better feature representations. The key strength of Xception lies in its ability to extract both low-level and complex features through its deeply separable convolutions, making it ideal for tasks like disease classification in coconut tree leaves, where details such as intricate vein structures need to be detected.

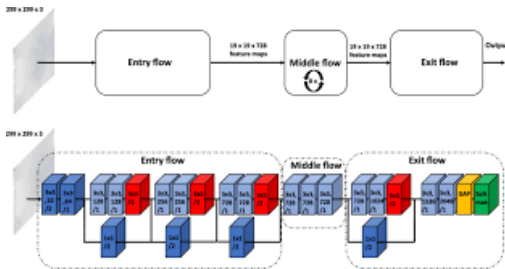


Fig. 3. Xception Architecture

3) *Ensembled model*: The 2 base models were selected on the basis of test evaluations which are performed for 6 different deep learning architectures of different weights and layers. They were run and based on their test scores in the classification report, VGG16 and Xception were chosen. The VGG 16 architecture has layers whereas the Xception architecture has 71 layers. These 2 models are ensembled by concatenating their GlobalAveragePooling scores together. They are then put into our customised neural network instead of deep learning algorithms and run using evaluation metrics like adams optimizer, accuracy etc.

IV. RESULTS

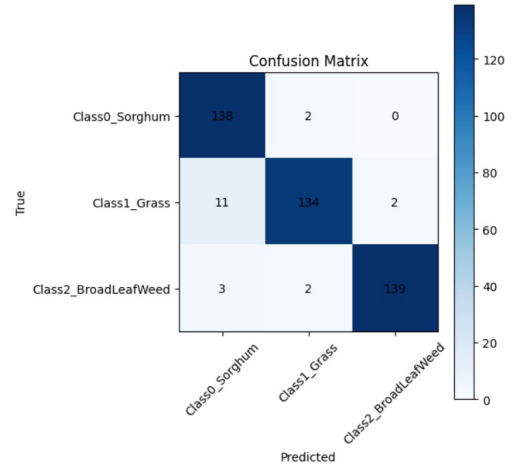


Fig. 4. Confusion Matrix Of Ensembled model

The diagonal elements of the confusion matrix are the correctly classified number of instances in each class. For example, 138 sorghum samples were correctly classified as sorghum, 134 grass samples were correctly classified as grasses, and 139 broadleaf grass samples were correctly classified as broadleaf grasses.

The diagonal interval elements represent the misclassified number of observations. For example, 11 grass samples were incorrectly classified as sorghum, 2 broadleaf grass samples were incorrectly classified as sorghum, 3 broadleaf grass samples were incorrectly classified as sorghum, were classified 2 sorghum samples were incorrectly classified as grasses, and 2 broadleaf grass samples were incorrectly classified as grasses.

Overall, this confusion matrix shows that the model has a high accuracy for each class. In each category, it correctly classified the majority of samples and made very few misclassifications. The performance of the model can be further evaluated by calculating metrics such as precision, recall, and F1-score for each class.

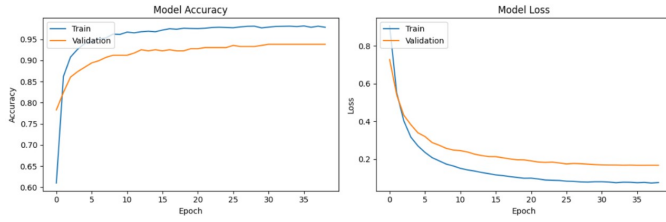


Fig. 5. Model Accuracy & Loss Of Ensembled model

The given figure illustrates the working of the training algorithm of the machine learning model. The left graph shows the accuracy of the model on the training and validation datasets over 35 epochs, while the right figure shows the corresponding loss.

The accuracy graph shows good quality, because the model achieves consistently high accuracy during the training phase. The training accuracy climbs rapidly at first and eventually plateaus around 0.96, indicating the success of the learning process. The validation accuracy exhibits a similar upward trajectory, reaching a plateau around 0.94, albeit slightly lower than the training accuracy. This observation indicates that the model generalizes well to the unseen data.

Conversely, the loss plot exhibits a downward trend as the model improves. The training and validation loss curves are consistently lower, indicating that the model is learning better from the data and its ability to predict outcomes is improving with each period. The validation loss curve still matches the training loss curve well, indicating that the model does not show significant overfitting, a common issue in machine learning where models perform well on training data but poorly on data they don't see it.

Overall, the figures show a well-trained model with good generalizability, indicating that the model learns well from data and provides accurate predictions. A small difference in accuracy and loss between training and between validation sets may prove too valid.

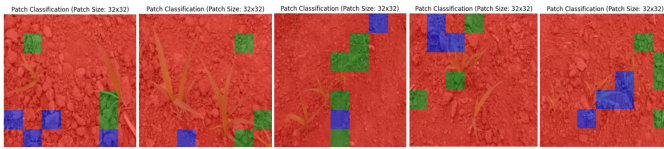


Fig. 6. Sorghum (Plant)

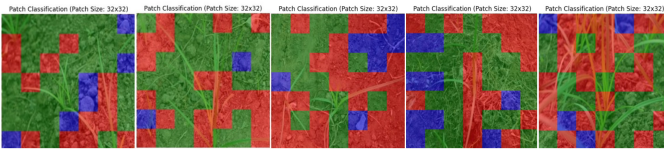


Fig. 7. Grass (weed)

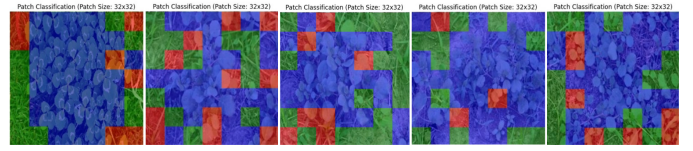


Fig. 8. Broad Leaf (weed)

The output images presented are the patch-wise analysed test images for the SORGHUM dataset. The sorghum plants are coloured red while the grass weeds and broadleaf weeds are coloured in green and blue respectively. The patch sizes were initially dynamically reduced from 64x64 to 8x8 based on uni-class or multi-class detection of weed species as explained earlier in the introduction. Then we changed the dynamic patch size allocation to static allocation, since the detection test performed were inconsistent in their performance.

Now, the patch size is iteratively run and optimal size is selected in trial and error process. In the input layer of the model leaky Relu activation function is used which was also selected in the same trial and error process among Relu, swish and Elu. The result contains 5 images of each classes and their patch-wise analysed and plotted.

V. CONCLUSION

In this study, an improved image classification and detection algorithm was developed for the SORGHUM weed data set, using the capabilities of two convolutional neural network architectures: VGG16 and Xception combined approach improved the classification accuracy especially on in the desert. The introduction of band-wise classification techniques has enabled the spatially accurate identification of grass species such as sorghum grass and broadleaf grasses through dynamic grid-based segmentation. Dynamic patch size allocation was used initially but later modified to static allocation for better organization in search performance. A trial-and-error approach with an optimized patch size and activation function selection, including the use of Leaky ReLU, contributed to the accuracy of the design.

Evaluation metrics including confusion matrices, classification reports, patch accuracy, class probability, and class-wise area contribution demonstrated the effectiveness of the patch-wise classification method in weed detection in agriculture. supporting the conclusion that the proposed method is effective in identifying and classifying sorghum and weed species at different granularity. This work contributes to the continued development of a comprehensive local agricultural weed management system through in-depth studies, and paves the way for improvements in crop management and weed control in precision agriculture.

VI. REFERENCES

1. X. Jin, J. Che, and Y. Cheng, "Weed identification using deep learning and image processing in vegetable plantation," College of Mechanical and Electronic Engineering, Nanjing Forestry University, Nanjing 210037, vol. 8, no. 1S4, pp. 1-6, Jan. 2021.
2. R. Gerhards, D. A. Sanchez, and P. Hamouz, "Advances in site-specific weed management in agriculture," Weed Science Department, University of Hohenheim, Stuttgart, Germany, Feb. 2022.
3. P. Rasappan and B. S. P. Balamurugan, "Computer vision and deep learning-enabled weed detection model for precision agriculture," M. Kumarasamy College of Engineering, Karur, Tamil Nadu, India, Aug. 2022.
4. M. Qi and H. Gao, "Method for segmentation of bean crop and weeds based on improved UperNet," Electrical and Information Engineering College, Jilin Agricultural Science and Technology University, Jilin 132101, China, Dec. 2023.
5. M. Li and S. Cheng, "High-performance plant pest and disease detection based on model ensemble with inception module and cluster algorithm," College of Plant Protection, China Agricultural University, Beijing 100083, China*, Jan. 2023.
6. M. T. R. Mahesh and V. K. Vinoth, "Early predictive model for detection of plant leaf diseases using MobileNetV2 architecture," International Journal of Intelligent Systems and Applications in Engineering, Feb. 2023.
7. F. D. Adhinata and R. Sumiharto, "A comprehensive survey on weed and crop classification using machine learning and deep learning," Artificial Intelligence in Agriculture, 2024, Elsevier.
8. A. S. M. Mahmudul Hasan, F. Sohel, D. Diepeveen, H. Laga, and M. G. K. Jones, "A survey of deep learning techniques for weed detection from images," Information Technology, Murdoch University, Murdoch, WA 6150, Australia and Centre for Crop and Food Innovation, Food Futures Institute, Murdoch University, Murdoch, WA 6150, Australia, Mar. 2022.
9. D. Hirata and N. Takahashi, "Ensemble learning in CNN augmented with fully connected subnetworks," IEICE Transactions on Information and Systems, vol. E106-D, no. 1, pp. 7-13, Jul. 2023, ISSN: 1745-1361.
10. X. Zhao, M. Zhang, and J. Zhang, "Ensemble learning-based CNN for textile fabric defects classification," International Journal of Clothing Science and Technology, vol. 33, no. 1, pp. 1-10, Jan. 2021, ISSN: 0955-6222.